BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Covalent Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. A shared pair of electrons.
2. An outer-shell pair not used in bonding.
3. e.g. H_2O or CO_2; diamond, graphite or SiO_2.

B

4. Each Cl shares one electron; the shared pair is attracted to both nuclei.
5. Each C bonded to four others in a tetrahedral giant covalent network.
6. Graphite has delocalised electrons between layers; diamond's electrons are all in C–C bonds.

C

7. 1, 2, 3 and 4.
8. Water is simple molecular (weak intermolecular forces). SiO_2 is giant covalent (many strong bonds must break).
9. Boiling overcomes intermolecular forces; covalent bonds inside molecules remain.

D

10. Each H has 1 electron. They share a pair. Both then have a full first shell. The pair is attracted to both nuclei.
11. Iodine is simple molecular — weak forces between I_2 molecules. Diamond is giant covalent — many strong C–C bonds must break.
12. No mobile ions or delocalised electrons.

E

13. Nitrogen has a lone pair. The pair is donated into a vacant orbital on H^+ to form a dative bond. All four N–H bonds in NH_4^+ are equivalent once formed.